# STAT 330-001

40142034 27

## Sommer, Henry Porter

784608753 sommerh0 385-209-7428

Test: 639340 Form: A
Entry: 08:32am 10/21/2026
Employee: Annabelle H.

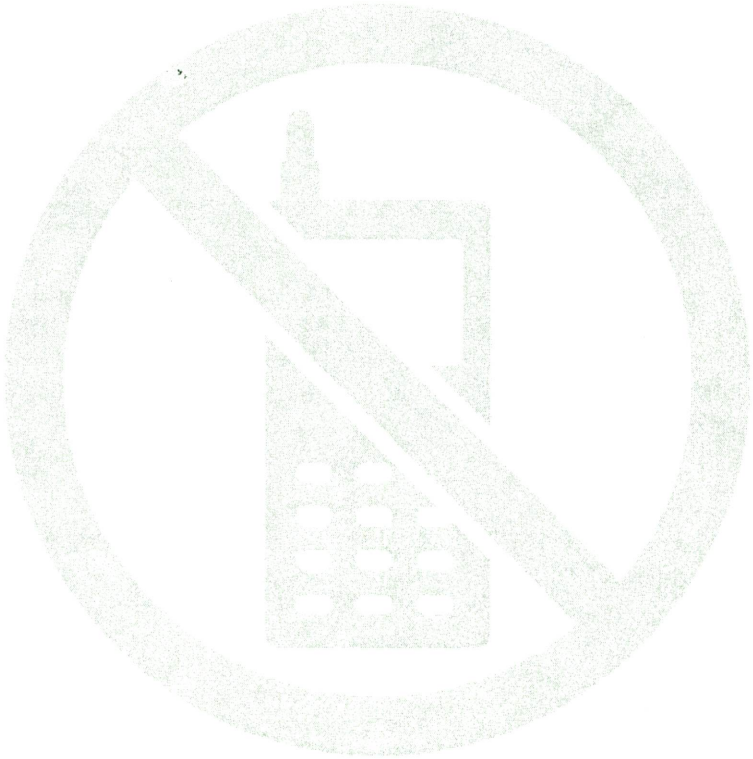
Wadsworth, Spencer Gordon
Checked out:

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

## PLEASE TURN OFF YOUR CELL PHONE



NS

40142034

D27

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1:** **(Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2:** **(TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3:** **(Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4:** **(Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5:** **(Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6:** **(Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7:** **(Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They are reffered to as "shrinkage" methods because they shrink or move our $\hat{\beta}$s closer to 0. This is done by having a penalty with $\lambda$ that encourages more stable and moderate coefficients.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right). \quad \left(\frac{2.4^2}{4}\right)\left(\frac{0.25}{0.75}\right) = 0.48$$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\hat{\beta}$?

$$\hat{\beta} = (X^TX)^{-1}X^Ty$$

(B) What is the conceptual difference between $\hat{\beta}$ and $\beta$?

$\beta$ is the true population coefficients. $\hat{\beta}$ is our sample coefficients that we calculated by minimizing the SSE.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

we know $SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ this is $SSE = (y - X\beta)^T(y - X\beta)$

(1) taking transpose of second term we can combine 2nd & 3rd

$SSE = y^Ty - y^TX\beta - \beta^TX^Ty + \beta^TX^TX\beta$

(3) then we distribute to get

$SSE = y^Ty - 2\beta^TX^Ty + \beta^TX^TX\beta$

(2) now we take gradient in terms of $\beta$

(4) when we take the gradient we set $SSE = 0$ & change $\beta$ to $\hat{\beta}$. If we solve for $\hat{\beta}$ we will find the $\hat{\beta}$ that minimizes SSE.

$0 = -2X^Ty + 2X^TX\hat{\beta}$

(5) $X^Ty = X^TX\hat{\beta}$ ← normal equations

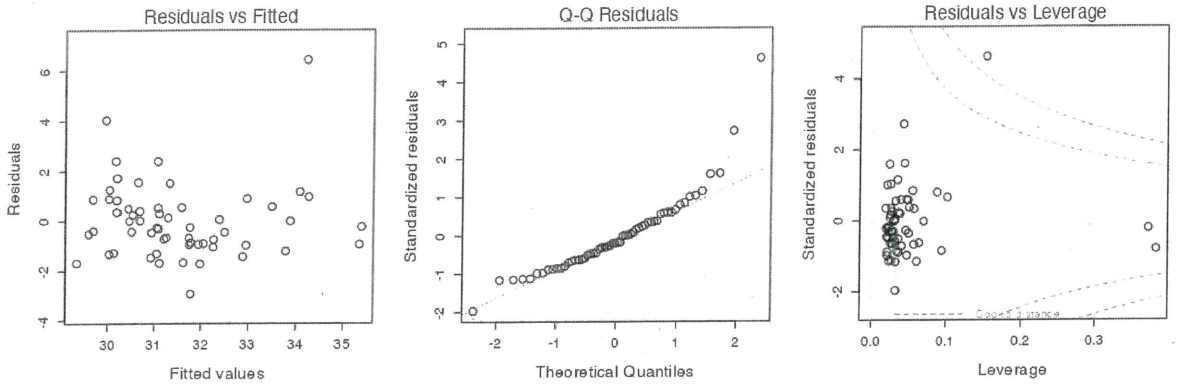
(6) so last we solve for $\hat{\beta}$ and get $\hat{\beta} = (X^TX)^{-1}X^Ty$ This $\hat{\beta}$ will be the vector of $\hat{\beta}$s that yield

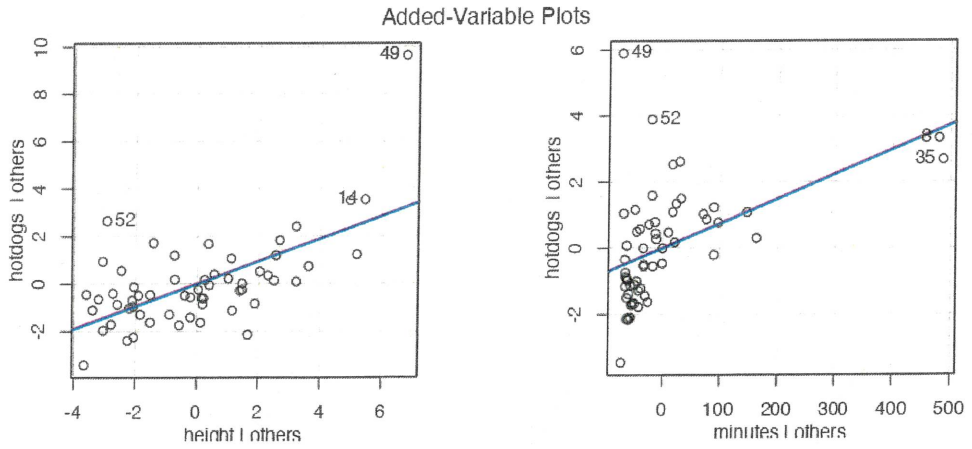
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

The estimated value of $\beta_1$ is $\hat{\beta}_1 = 0.4729$. Holding # minutes trained per week constant, a one inch increase in height is associated with a 0.4729 increase in # of hotdogs consumed in the 10 minute sitting.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Looking at the residuals vs fitted plot, I see no obvious patterns. The added variable plot for height given others also shows little concern. The minutes|others is a little strange but again I see no obvious pattern. So I would say the data satisfy normality fairly well.

(B) There is independence between each observation

There was not a random sample of people taken to compete. Some eaters may be relatives or be connected in some way that could make independence between observations be questionable. I would say the independence assumption is not well satisfied, but we can still continue.

(C) The errors are normally distributed

The shapiro-wilk p-value is 2.43e-8. This means we have evidence to assume the residuals do not come from a normal distribution. The residuals also do not stay very well on the theoritical quantile line of the QQ plot towards the right. I would say the data do not satisfy the normality assumption very well. But just because the residuals do not come from a perfectly normal distribution, it doesn't mean that that distribution isn't close to normal.

(D) The errors have constant variance (homoscedasticity)

With the p-value of the Breusch-Pagan test at 0.071, we don't have much evidence to assume the error variance depends on our predictors. Based on that (and no funnel shape on Residuals vs. Fitted plot) I would say the equal variance assumption is fairly well satisfied.

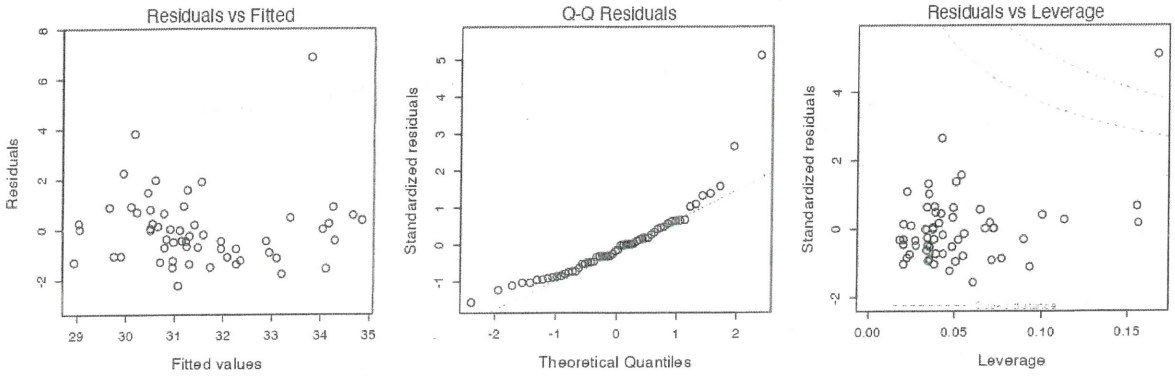
(E) There are no unduly influential points

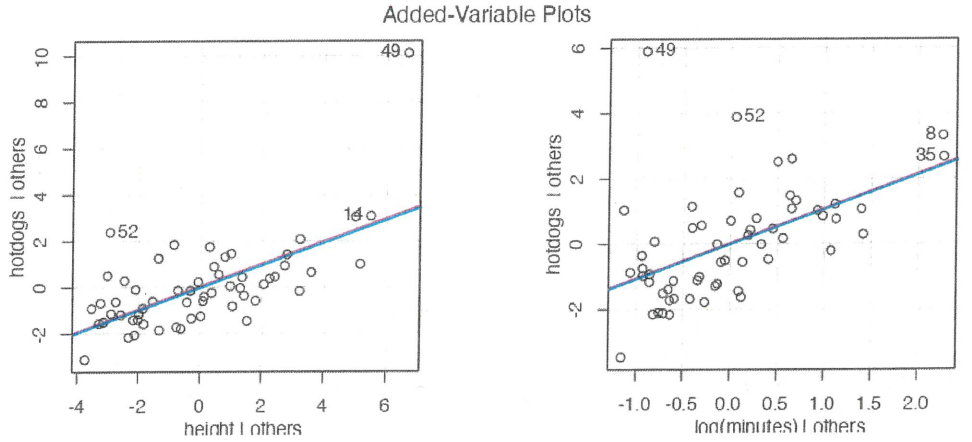
In the residuals vs leverage plot, there is one observation with a high enough cook's distance to be beyond a certain threashold that R has identified with a dotted line. We would definetly want to examine this observation before assuming that we be satisfied with this assumption/check.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

The breusch-pagan test is just used to measure if the error variance is constant or if it depends on the predictors. Although model 1 has a more promising p-value for that test, the test is only one part of many different tests & plots that determine if a model is useful.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages: Model 2 has smaller p-values for intercept & both predictors. It also has a higher shapiro-wilk test p-value & I lower CV RMSE. The plots are similar but it's added-variable plot for log(minutes) | others looks a little better. (More evenly spreaded points)

Disadvantages: Model two has a lower breusch-pagan p-value & the influencial observation (with cooks distance) from model 1 looks even more influencial after the transformation in model 2.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\hat{\beta}_2 = 1.067$. Holding height constant, a 1% increase in minutes of training per week is associated with an estimated change of about $1.067/100 = 0.01067$ units in the mean of hotdogs eaten. The exact estimated increase is $1.067 \log(1 + \frac{1}{100}) = 0.010617$ per 1% increase in minutes of training per week

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\widehat{hotdogs} = -6.056 + 0.488(height) + 1.067(log(minutes))$

$-6.056 + 0.488(68) + 1.067(4.369)$

$= 31.790$ hotdogs

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

$df = n - p - 1$

$58 - 2 - 1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$CI = estimate \pm t^* \cdot SE$

we use the $s\sqrt{1 + x^*(X^\top X)^{-1}x^*}$ for a prediction interval

so $31.790 \pm 2.004 \cdot 1.485$

$31.79 \pm 2.97594$

$CI = (28.814, 34.766)$

We are 95% confident that the true amount of hotdogs my friend will be able to eat in the 10 minute sitting is between 28.814 and 34.766 hotdogs.

# Part III - Application

## Questions 20-24 concern the following dataset:

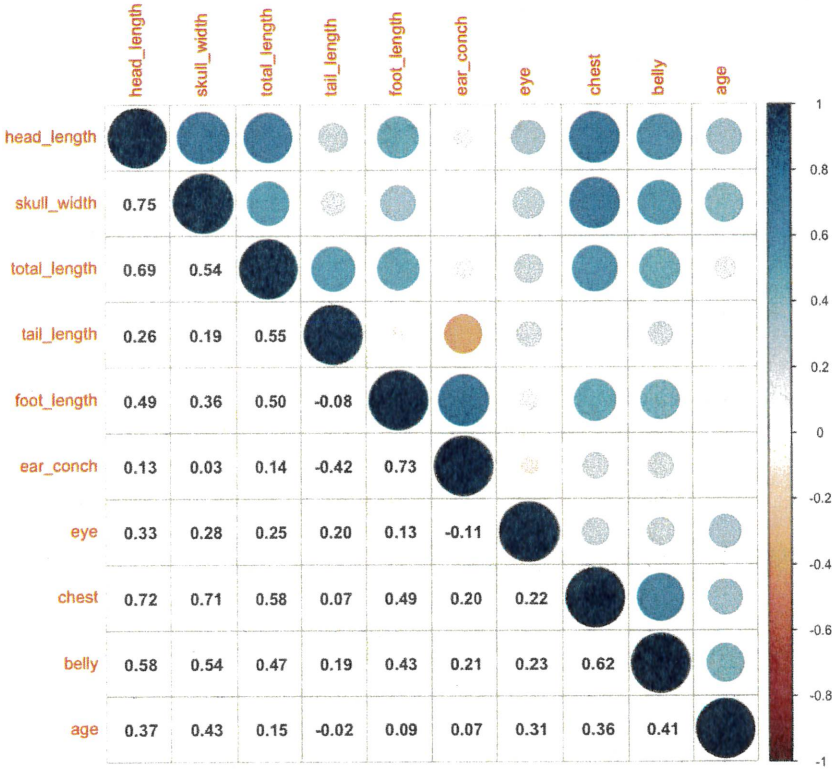
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

Coefficient 0.1655a $t^* = 2.01$

SE = 0.1059a

$0.1655a \pm 2.01 \cdot .1059a$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true relationship between belly width & age (holding all other predictors constant) is between -0.023 & 0.443.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Even if there isn't an extremely high probability (low p-value) that any one predictor is significant, there can be a very high probability (low value) that at least one is significant, which is what the F-test evaluates.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best subsets is a model search algorithm that goes through every possible model. AIC is a model selection metric.

So best subsets would go through every model and we would use AIC to determine which of those models we prefer based on which one gives the best score by using AIC's penalty.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 26

## Lopez Paredes, Maria Fernanda

279050379 mariafl 385-452-1868

Test: 639072 Form: A Wadsworth, Spencer Gordon

Entry: 07:41pm 10/20/2026 Checked out:

Employee: Elijah B.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

Fernanda Lopez

NS

40142034

026

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

LASSO and ridge called shrinkage methods because they make regression coefficients smaller, closer to zero. They do this by adding a penalty to the model. LASSO can make some coefficients exactly zero, while ridge makes them smaller but usually not zero.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$$\widehat{\beta} = (X^T X)^{-1} X^T y$$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\beta$=represents the true population coefficients
$\widehat{\beta}$=represents the estimated coefficients calculated from sample data

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

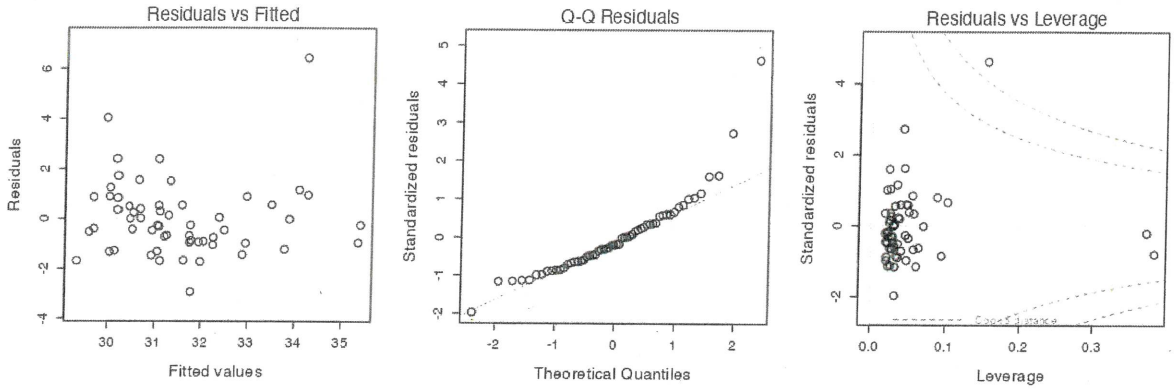
We use least squares to find the best fitting regression line, it adjusts the coefficients until the squared distances between the data points and the model's predictions are as small as possible.

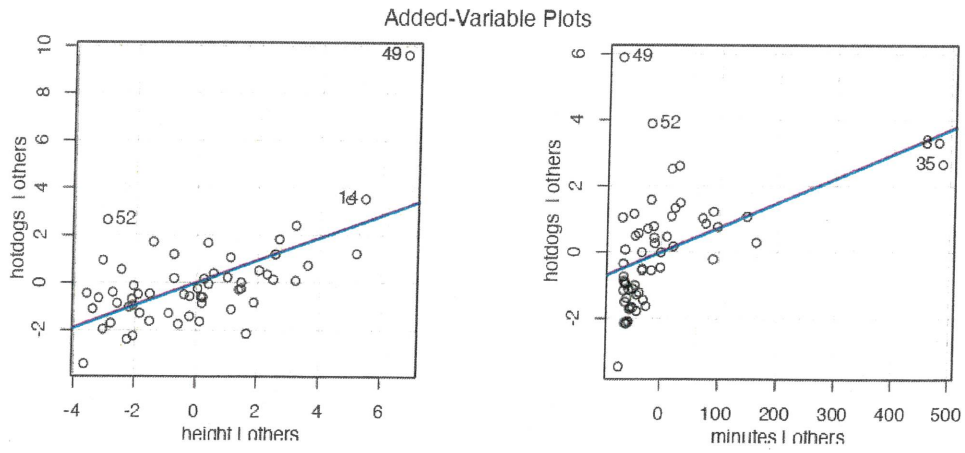
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\hat{\beta}_1 = 0.472892$

For each additional inch of height, the number of hotdogs eaten increases by aprox 0.472892

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The residual plots suggest some nonlinearity, specially for training minutes. This assumption may not be satisfied.

(B) There is independence between each observation

Each contestant was measured separately, but independence cannot be confirmed from the plot.

(C) The errors are normally distributed

The shapiro wilk p value is $2.43 \times 10^{-8}$, which is less than 0.05. The normality assumption is violated.

(D) The errors have constant variance (homoscedasticity)

The Breush Pagan p valle is 0.071 is greater than 0.05

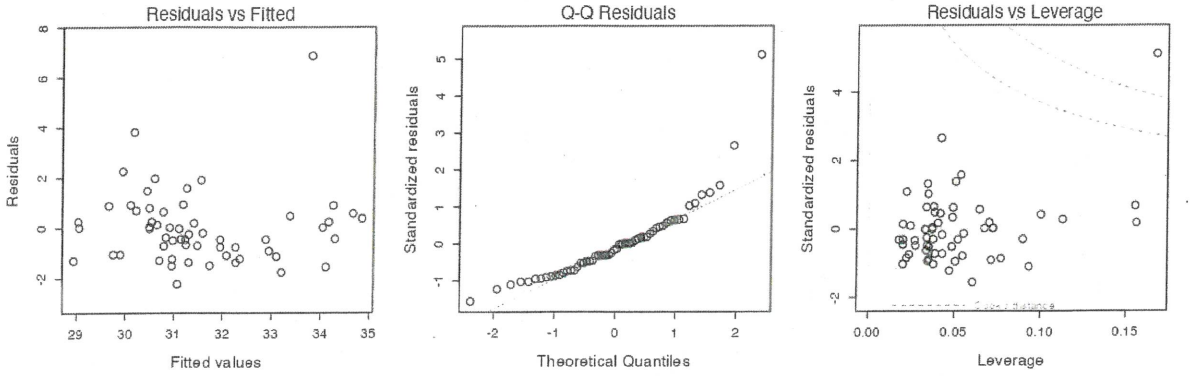
(E) There are no unduly influential points

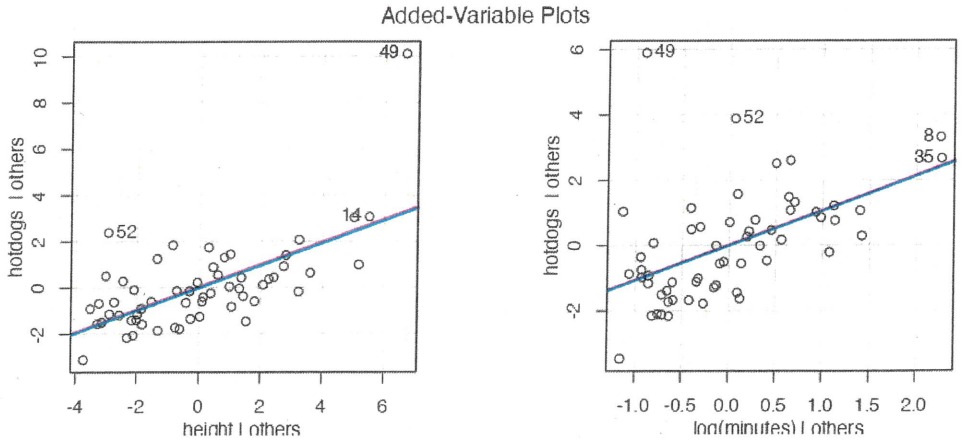
The plots show some points with height leverage, so influential observations maybe present. More investigation is needed.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| `log(minutes)` | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I desagree
Model 1 has less evidence of unequal variance (p=0.071 vs 0.028) However we should also compared prediction accuracy and other model assumptions before deciding which model is better

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 has better prediction accuracy because its CVRMSE is lower (1.66702 vs 1.84623)
The log transformation may also improve linearity. However model 2 has more evidence of unequal variance (p = 0.028) and still has non normal residuals.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\hat{\beta}_2 = 1.06702$
Doubling weekly traing minutes increases approx .74 Hotdogs eaten

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$-6.05589 + 0.48834(68) + 1.06702(4.369)$

$= 31.81$ Hotdogs

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.81 \pm 2.976 = (28.83, 34.79)$

# Part III - Application

## Questions 20-24 concern the following dataset:

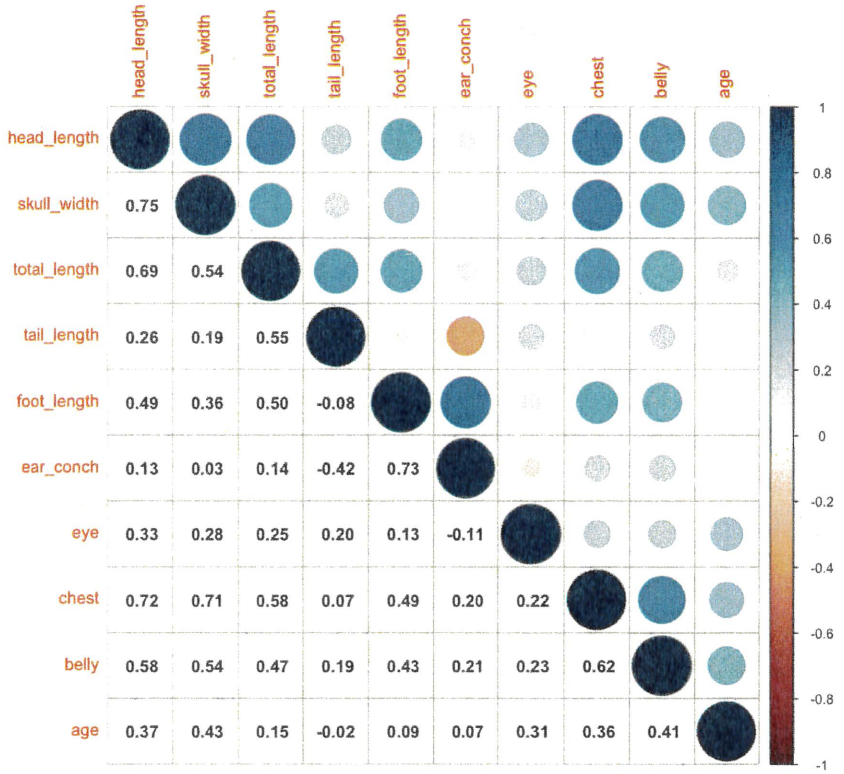
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 0.21304$

$= (-0.04745, 0.37863)$

$(-0.04745, 0.37863)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that each unit increase in belly width changes possum age by -0.023 to 0.443 years, holding other variables constant. since interval includes 0, the effect is not significant.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

multycollinearity is likely causing this the predictors are correlated, making individual effects harder to identify. together, they can still predict age

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The algorithm tries every possible group of variables and uses AIC to find the best model. The one with the smallest AIC is chosen.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:

```
lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)
```

This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 25

## Keep, Mason Hawthorne

907462645 mkeep 805-591-0379

Test: 638589 Form: A Wadsworth, Spencer Gordon

Entry: 05:52pm 10/20/2026 Checked out:

Employee: Maya N.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

40142034

025

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE The model itself stays the same, but we account for other variable to try to isolate its effects.

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated. not necessarily

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Our coefficient estimates ($\hat{\beta}_i$) are shrunk. This is accomplished by adding a penalty term ($\lambda\sum\beta_i^2$ or $\lambda\sum|\beta_i|$) to the term $\sum_{i=1}^{n}(y_i-\beta_0-x_i^T\beta)^2$ and finding values of $\hat{\beta}_i$ that minimize this sum.

$4/n = 0.04$

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right) = \left(\frac{2.4^2}{4}\right)\left(\frac{0.25}{0.75}\right) = 0.48$$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\beta$ is the population parameter for the coefficients in our MLR model, $\hat{\beta}$ is our sample statistic and our estimates for $\beta$. for the coefficients

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

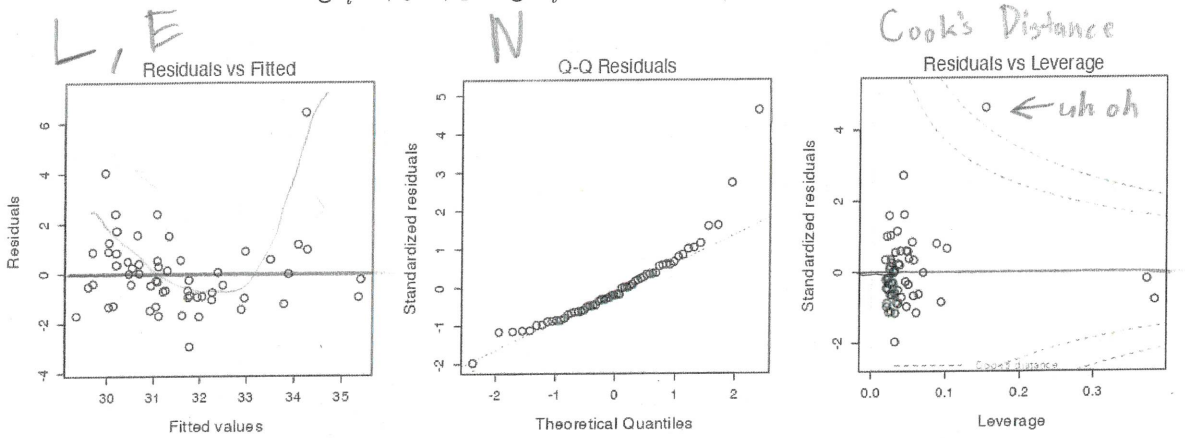
Least squares chooses $\hat{\beta}$ that minimizes $(Y-X\beta)^T(Y-X\beta)$. To do this, we would distribute this product and simplify. We would then take the gradient of the expression and set it equal to zero (this finds the minimum value for $\hat{\beta}$). After simplifying, this process yields the normal equations $X^TX\hat{\beta} = X^TY$; that is, $\hat{\beta} = (X^TX)^{-1}X^TY$.

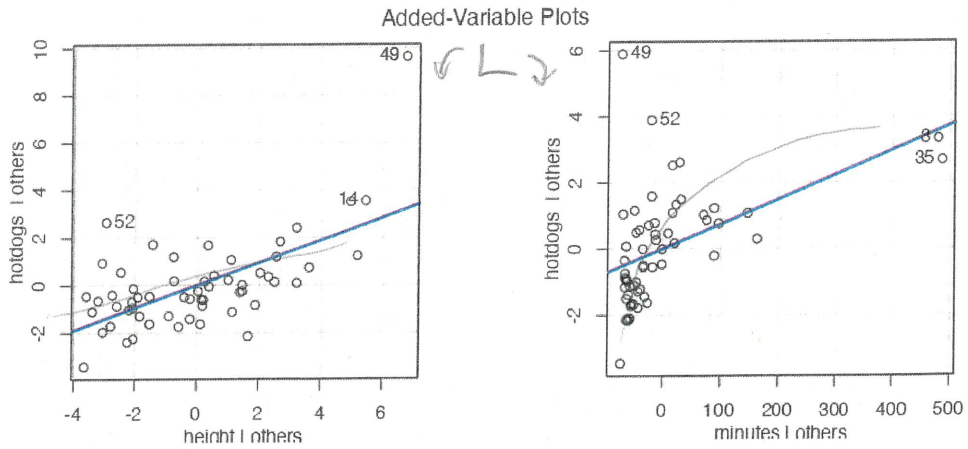
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\hat{\beta}_1 = 0.472892$

Holding the number of minutes trained constant, a one-inch increase in height is associated with a 0.472892 hotdog increase in total number of hotdogs eaten, on average.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Based on our residuals-versus-fitted plot, we have evidence to support the claim that the relationships between height/minutes and hotdogs are nonlinear. There seems to be a curve pattern in this plot. Based on our added-variable plots, we note similar evidence that the relationships between height/minutes and hotdogs are nonlinear (minutes vs hotdogs follows a curve and not a line, roughly).

(B) There is independence between each observation

Independence between each observation is likely a reasonable assumption. Each contestant's height is likely not influenced by the others' neither the minutes. One possible way independence could be violated is if county-wide competitions or workshops were held, increasing ^average # of total hotdogs for those in the county.

(C) The errors are normally distributed

Evidence from our Q-Q plot shows that our standardized residuals may be skewed right (not normally distributed). The Shapiro-Wilk test for normality rejects the null hypothesis that the errors are normally distributed (p-value of 2.46e-07) at the 0.05 significance level, supporting our analysis of the Q-Q plot that this assumption may not be reasonable.

(D) The errors have constant variance (homoscedasticity)

We have some evidence to support the claim that the errors have constant variance. Judging by our residuals-versus-fitted plot, there seems to be a slight funnel-shape. Our Breusch-Pagan test does not reject the hypothesis that we have constant variance (p-value of 0.071) at the 0.05 significance level, but this does not mean that this assumption is not slightly violated.

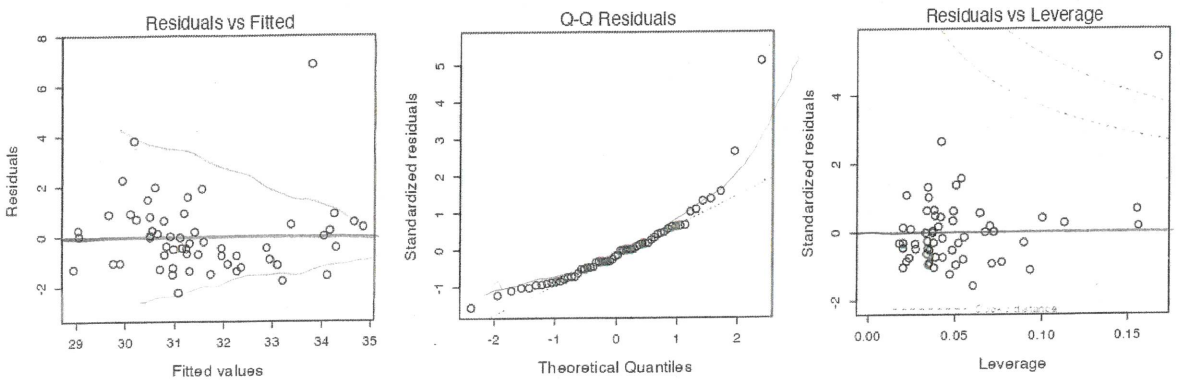
(E) There are no unduly influential points

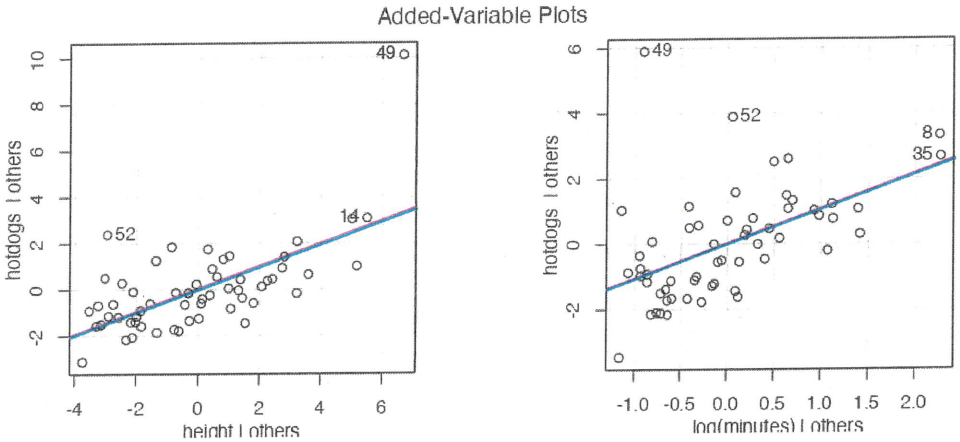
We have evidence to support the claim that there are unduly influential points based on our residuals-versus-leverage plot. One point has a high Cook's distance and is influential in model estimates.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

This model (1) is not clearly superior to model 2. Although perhaps the assumption of homoscedasticity is better met in model 1, that doesn't mean that the other assumptions are sufficiently satisfied better in model 1 versus model 2. p-values from tests are not clear-cut rules either, we should examine our data and plots to examine the extent of assumption-violation.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Some advantages model 2 provides over model 1 are ~~that~~ the we have more evidence supporting linearity (less of a curve-pattern in residuals-versus-fitted plot and more linear in our added-variable plot for minutes vs hotdogs). The assumption for normally-distributed errors seems to be less violated (slightly less of a curve in Q-Q plot and Shapiro-Wilk's p-value is slightly larger). This model seems to be able to predict outside data better as well (our cross-validated RMSE is lower in model 2, this holds well if the CV RMSE was computed using the same folds). Some disadvantages include a likely greater violation of our assumption of homoscedasticity (our residuals-versus-fitted plot shows a clearer funnel-shape and the p-value for our Breusch-Pagan test is lower). Another disadvantage is that our $\hat{\beta}_2$ is harder to interpret.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\hat{\beta}_2 = 1.06702$ $\quad \Delta \overline{hotdogs} = \hat{\beta}_2 \cdot \log(1.01) = 0.0106172$

Holding height constant, a 1 percent increase in trained minutes is associated with an average increase of 0.0106172 hotdogs in total hotdogs consumed, on average.

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\widehat{\text{hotdogs}} = -6.05589 + 0.48834(68) + 1.06702(\log(79)) = 31.8135$

The predicted number of hotdogs my friend can consume is about 31.81 hotdogs.

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$n - p - 1$
$= 58 - 2 - 1$
$= 55$

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

PI: $\widehat{\text{hotdogs}} \pm t^* \cdot s\sqrt{1 + x^*(X^\top X)^{-1}x^*}$
$= 31.8135 \pm (2.004)(1.485)$
$= 31.8135 \pm 2.97594$
$= (28.8376, 34.7894)$

We are 95 percent confident that my friend can consume between 28.8376 and 34.7894 hotdogs in 10 minutes.

# Part III - Application

## Questions 20-24 concern the following dataset:

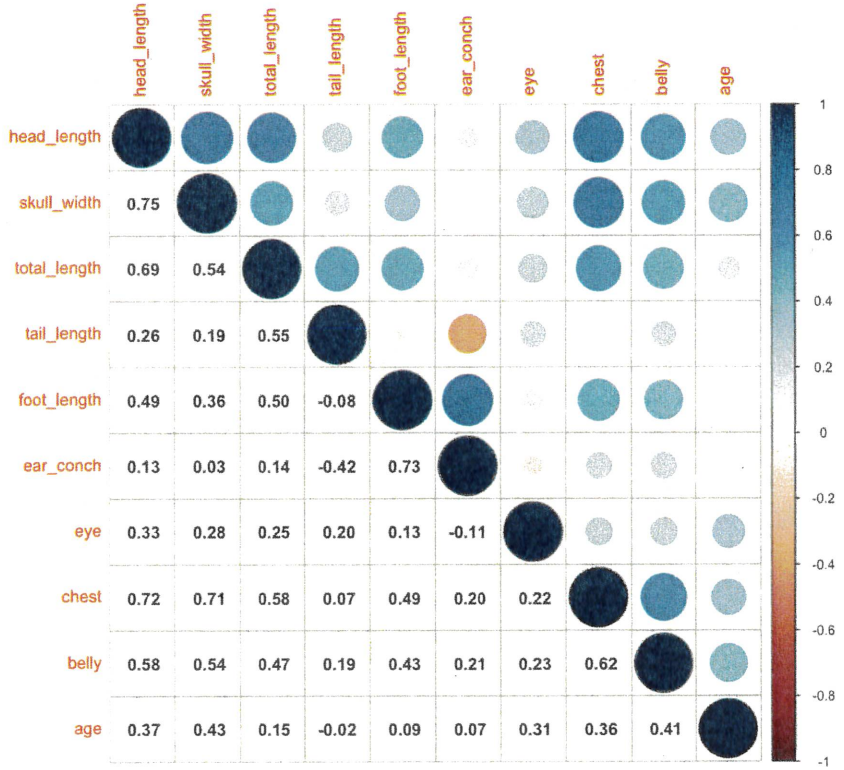
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$pt_est \pm t^* \cdot SE$

$0.16559 \pm 2.01 \cdot 1.0599$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95 percent confident that the true mean increase in age for a 1 cm increase in belly girth holding all other predictors constant is between -0.023 (representing a decrease of 0.023 years) and 0.443 years.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Multicollinearity. An average VIF of 3 may suggest our full model has problems with multicollinearity, which means that although our F-test may provide evidence that at least one $\beta$ is nonzero, our individual model coefficients don't have a significant effect ($\alpha = 0.05$) because the other variables may explain the estimated coefficient.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm using AIC as its criterion (or metric) goes through every possible model and computes AIC. AIC is equal to $n\log(SSE/n) + 2k$, where $n$ in this case is 59. $k$ is often $n-2$. After computing AIC for every possible model, the best subsets algorithm selects the model with the smallest AIC.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 24

## Hooley, Ethan Talmage

089533609 ethanhoo 435-317-3739

Test: 638571 Form: A Wadsworth, Spencer Gordon

Entry: 05:48pm 10/20/2026 Checked out:

Employee: Maya N.

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

40142034

024

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are-allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1:** **(Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE
(B) FALSE

**Question 2:** **(TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE — log transformation
- FALSE

**Question 3:** **(Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

adj can decrease but normal $R^2$ can only ↑ or stay same

(A) always increases or remains unchanged.
(B) always decreases or remains unchanged.
(C) may increase or decrease depending of the additional variable.

**Question 4:** **(Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

= high VIF

(A) It can cause the linearity assumption to be violated.
(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.
(C) It can result in larger standard errors, and thus greater uncertainty in estimates.
(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS more bias

(C) They produce more accurate confidence intervals and hypothesis tests ↓ variance

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

The predictor variables are being shrunk towards 0 (and with lasso sometimes to 0) by increasing bias but in turn decreasing variance.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$D_i = \frac{2.4}{4} \times \left(\frac{0.25}{1-0.25}\right) = 0.6 \times 0.33 = 0.198 \approx 0.2$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

$D > 4/n$ interesting

$0.2 > 0.04$

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^T X)^{-1} X^T y$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\hat{\beta}$ is the estimated coefficient while the $\beta$ is the true value

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

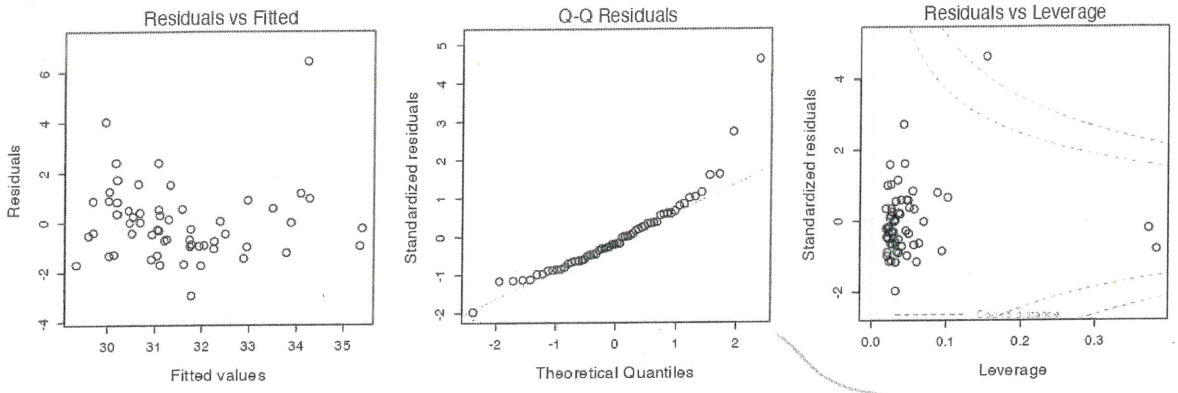
By using least squares we can estimate our regression coefficients by plotting our data points and seeing the effect the estimate has on the response variable.

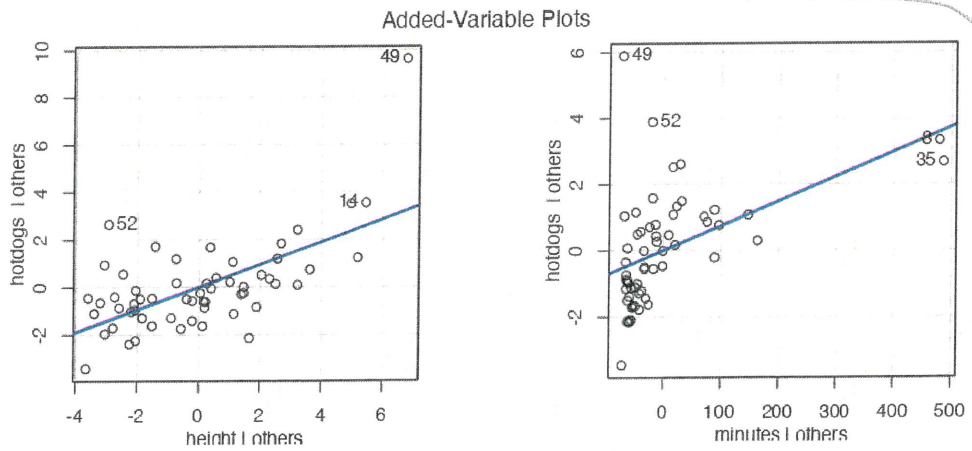
# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

$\beta_1$ height = 0.472892, while holding minutes coefficient constant, for each increase in 1 in. on height, there was a positive increase of 0.472892 for the response variable which is number of hot dogs eaten.

**Question 12:** **(Circle all that apply)** Which of the following are true? $n=58$ $t^*=?$

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$ $\hat{\beta} = .4729$

$.007417 \pm 3.963(.001871)$

$(.00000223, .01483)$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Satisfy linearity assumption. Resid. vs. Fitted looks like you can draw a straight line through while added variable plots show no curvature. Maybe slight curvature on Resid vs. Fit. plot.

(B) There is independence between each observation

Might have problems. Each observation can be influenced by the last. As someone consumes more hot dogs, the harder it gets to eat more hot dogs.

(C) The errors are normally distributed

Assumption not met. Skewed tail Q-Q plot and Shapiro-Wilk p-value is significantly smaller than $\alpha = 0.05$ suggesting errors are not normally distributed. Reject null of normality of errors.

(D) The errors have constant variance (homoscedasticity)

Br.-Pag. p-value $> 0.05$ so we fail to reject null hypothesis of homoscedasticity. (No scale-location plot)

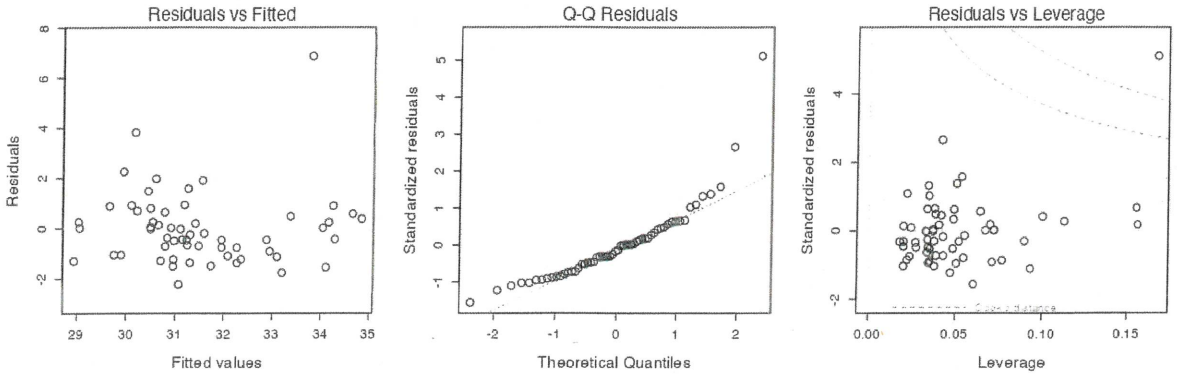
(E) There are no unduly influential points

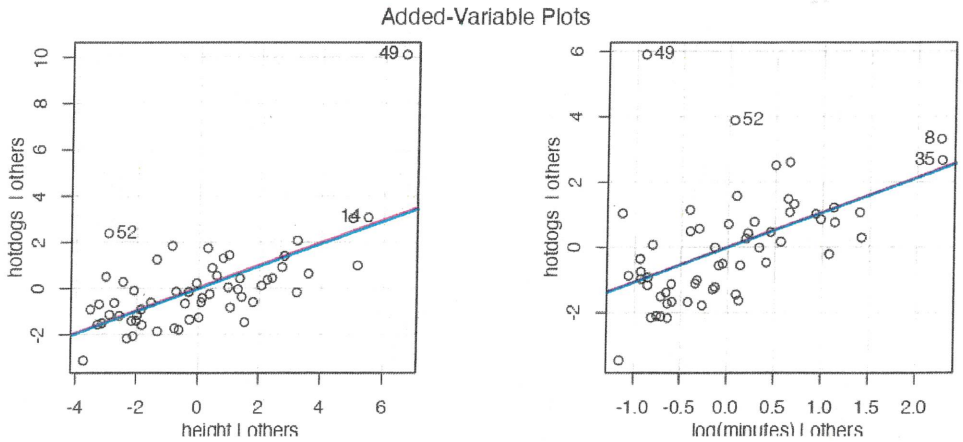
~~Hard to tell because~~ Based on resid. vs. Lev. plot there may be points with high influence as the have high leverage and high residuals. Need more info (such as Cook's D) to confirm level of influence that the points have on the data. Also checking regression line with and without points can help show whether or not these points

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would say a p-val of .028 < a of 0.05 suggesting unequal variance in the data, however, you cannot base a model's superiority on the equal variance assumption alone.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages of model 2 are that it has a lower CV error which means it is slightly better at making predictions on outside data. It also seems to have fewer points with a high leverage potentially decreasing the error that comes with heavily influencial points. I believe the log transformation

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\hat{\beta}_2 = 1.06702$. While holding height constant, for every increase in minute (training), there is a 1.06% increase on the response variable which is the number of hotdogs eaten.

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions. $\ln(10) = 2.3$

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\text{hotdogs} = \underbrace{-6.06 + (0.49)(68)}_{27.26} + (1.07)(\log(10)) + \varepsilon$

$+$

$\approx 27.26$ hot dogs

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

$n - p - 1 = 58 - 2 - 1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$1.485 \pm 2.004(.204) = (1.08, 1.89)$

$1.0672 \pm 2.004(1.485) = (-1.908, 4.024)$

↓ exponentiate $(e^{\#})$

$(0.148, 56.826)$

# Part III - Application

## Questions 20-24 concern the following dataset:

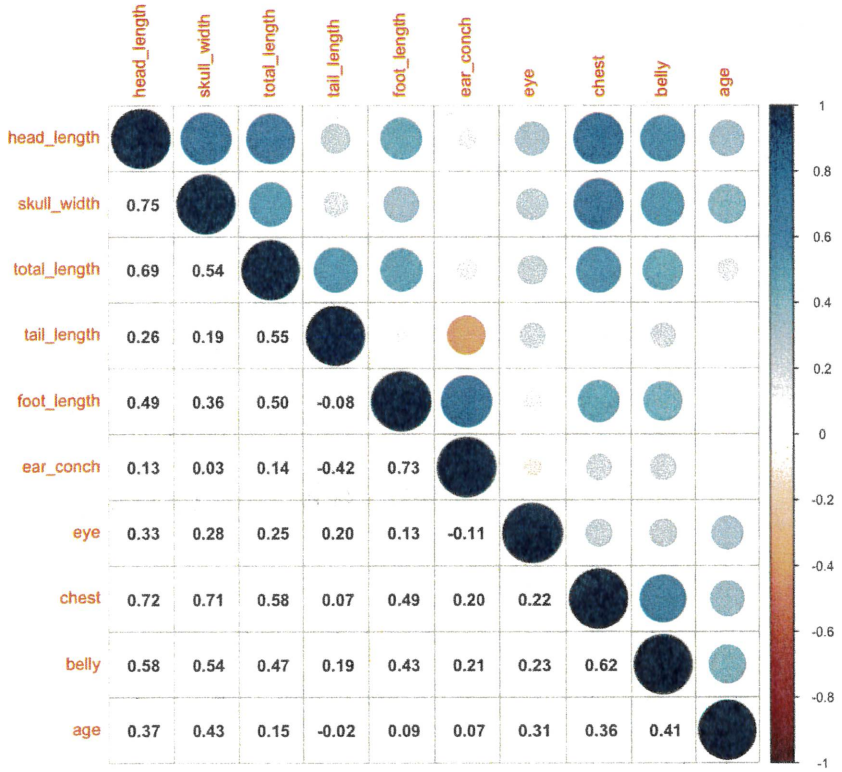
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01(.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident the true mean for $\beta_{belly}$ lies within the interval $(-0.023, 0.443)$

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Most likely the is multicollinearity. The small p-value allows us to reject the null hypothesis that there are no predictor variables with a nonzero relationship to the response.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Instead of adding and checking one extra variable at a time, best subsets checks all combinations of predictor variables and their relation to the response while searching for a model that can minimize the AIC value without having too much of a penalty on the predictors.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

full model vs. model chosen

find ↓ CV RMSE

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

split → fit train → test RMSE → repeat for loop

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

A Average RMSE across all test-training splits and report

C Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

C Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

C Compute each model's AIC and BIC

D Fit each model using the testing data

A Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

C Use the models fitted to the training data to get predictions for the test data

END OF EXAM

# STAT 330-001

40142034 23

## Paxton, Davis

030460122 dbod911 808-838-9626

Test: 638563 Form: A
Entry: 05:46pm 10/20/2026
Employee: Dylan S.

Wadsworth, Spencer Gordon
Checked out:

**SPECIAL INSTRUCTIONS**

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

**PLEASE TURN OFF YOUR CELL PHONE**

# STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

**Instructions**

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

## PART I

**Question 1: (Circle one answer)** Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

**Question 2: (TRUE/FALSE)** If the relationship between $x$ and $y$ is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

**Question 3: (Circle one answer)** When adding an additional variable to a linear regression model, the model's $R^2$ ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

**Question 4: (Circle all that apply)** Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination ($R^2$) to become negative.

**Question 5: (Circle all that apply)** Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the $x$'s or $y$ may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of $\beta$ can be obtained by fitting a simple linear regression for each $x_k$ only.

(E) The assumptions on the error variables (i.e., the $\epsilon_i$) in a multiple linear regression model are exactly the same as they are for simple linear regression.

**Question 6: (Circle one answer)** What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

**Question 7: (Circle all that apply)** In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

**Question 8:** Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Because in LASSO, the coefficients are shrunk and some can be set to 0, or essentially removed from the model, and Ridge also shrinks coefficients, although more smoothly and only towards zero. This is accomplished by either squaring the betas (Ridge) or taking the absolute value of the betas (Lasso) and putting a penalty (Lambda) on them, where higher lambda values shrink the betas more.

**Question 9:** A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation $i$ has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

**(Circle one answer)** Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

**Question 10:**

(A) What is the formula for $\widehat{\beta}$?

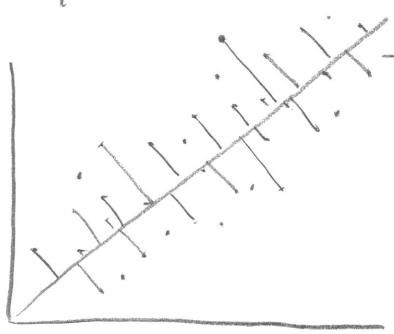
$(X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and $\beta$?

$\beta$ is a population parameter that is unknown, and $\widehat{\beta}$ is our fitted coefficient based on our particular sample, used to estimate the true $\beta$

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

Least squares chooses the coefficients (betas) that minimize the sum of the squared errors.



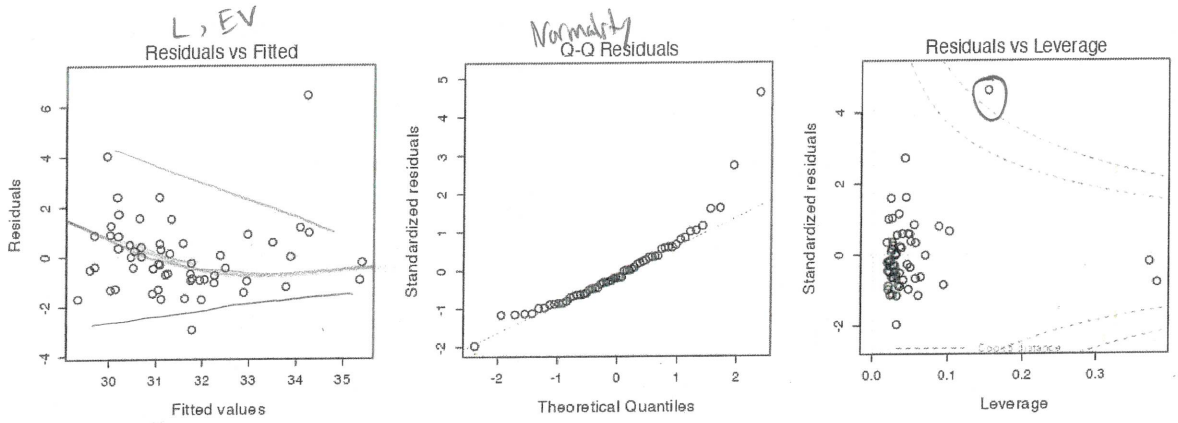
→ It chooses the slope that minimizes the sum of these squared errors

# PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

## Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$




| Coefficient | Estimate | Std. Error | $t$ value | Pr($>\|t\|$) |
|---|---|---|---|---|
| Intercept | -1.295392 | 5.467711 | -0.237 | 0.813602 |
| `height` | 0.472892 | 0.080352 | 5.885 | 2.46e-07 |
| `minutes` | 0.007417 | 0.001871 | 3.963 | 0.000215 |

EV
N

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.071 |
| Shapiro-Wilk p-value | 2.43e-8 |
| CV error | 1.84623 |

**Question 11:** State the estimated value of $\beta_1$, and interpret its value in the context of the problem

0.47.

While holding minutes of weekly training constant, a 1-inch increase in height is associated with a .47 increase in hot dogs consumed, on average.

**Question 12:** **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of $\beta_2$ does **not** contain 0

(B) The 95% confidence interval of $\beta_2$ does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

**Question 13:** Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The data does not satisfy the linearity assumption well. The residuals vs. fitted values shows a slight curve in the relationship, and specifically the added-variable plot of minutes also has moderate curvature

(B) There is independence between each observation

In general, independence between contestants, their heights, and minutes trained makes sense in the context of this problem. However, there could be an important grouping variable like county that may account for variation. Since they are clustered primarily into these counties, they are technically not independent but it may not practically matter for our question of interest

(C) The errors are normally distributed

The errors have a slight right skew, and a p val $< .05$ for the Shapiro Wilk test. However, the CLM is working in the sample's favor with $n = 58$, and the slight deviation from perfect normality most likely has no practical bearing.

(D) The errors have constant variance (homoscedasticity)

The errors do seem to have a systematic trend as the fitted vals increase. Even though the Breusch-Pagan test is marginally non-significant, the residuals vs. fitted shows the trend, and provides a hint at why the BP test is non-significant with a potentially high influence point in the top right

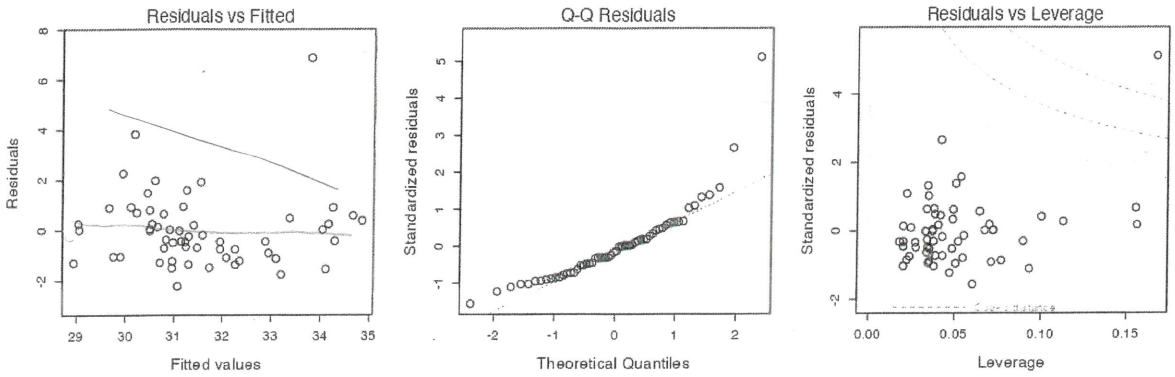
(E) There are no unduly influential points

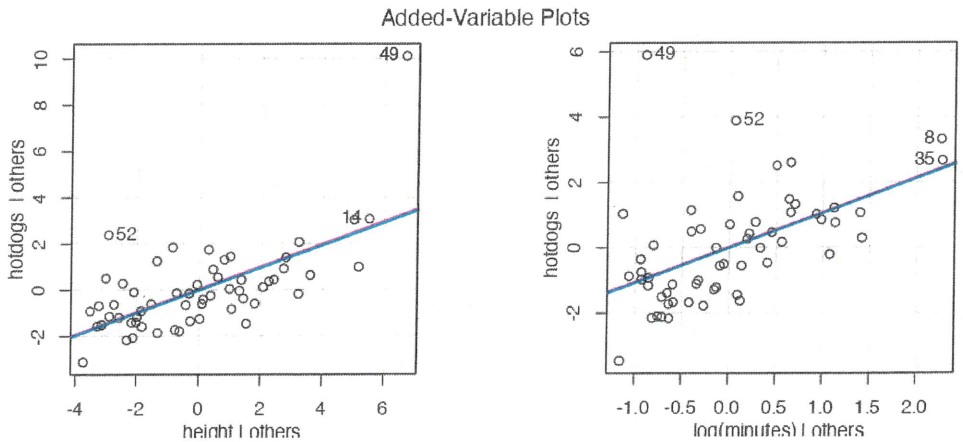
There are a couple points worth flagging for high leverage (strange) x-vals, but only one observation is truly worrisome with both high leverage (unique) and a standardized residual.

## Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$




| Coefficient | Estimate | Std. Error | $t$ value | $\Pr(>\|t\|)$ |
|---|---|---|---|---|
| Intercept | -6.05589 | 5.43620 | -1.114 | 0.27 |
| `height` | 0.48834 | 0.07821 | 6.244 | 6.48e-08 |
| log(`minutes`) | 1.06702 | 0.24079 | 4.431 | 4.51e-05 |

| Other Summaries | Value |
|---|---|
| Breusch-Pagan p-value | 0.028 |
| Shapiro-Wilk p-value | 3.43e-5 |
| CV RMSE | 1.66702 |

1.356

8.8, 7.9

**Question 14:** A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would say hold your horses mister. It depends on what you are prioritizing, and the tradeoff between satisfying better the equal variance assumption, or the linearity assumption. Since linearity matters for everything ($\beta$'s, C.I.'s, P.I.'s, F-tests), whereas slight violations of equal variance still have intact $\beta$ values, this transformation might make more sense despite potentially more unequal variance across the errors.

**Question 15:** Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

The main advantage is that it better satisfies our linearity assumption, making minutes spectrally more linear. However, it seems to have a tradeoff in having slightly larger CV RMSE (1.06 vs. 1.356), and more unequal variance of the errors.

**Question 16:** State the estimated value of $\beta_2$ and provide an interpretation in context

$\beta_2 = 1.06$

For every 1% increase in minutes practiced there is a .004 hot dog increase in the average hot dogs eaten.

101

OR

For every 1% increase in minutes practiced, the average hot dogs eaten increases by .004.

$1.06 \log(1 + 1/100)$

$1.06 \log(1.01) = .004$

## Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

**Question 17:** Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$-6.055 + .484(68) + 1.067 \log(79) = 31.79$

My friend is going to die.

**Question 18:** To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical $t^*$ used to calculate the margin of error?

59-2-1

(A) 55

(B) 57

(C) 56

(D) 48

**Question 19:** You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.79 + 2.004 \times (1.689)$

$31.79 - 2.004 \times (1.689)$

(28.406, 35.174)

(

# Part III - Application

## Questions 20-24 concern the following dataset:

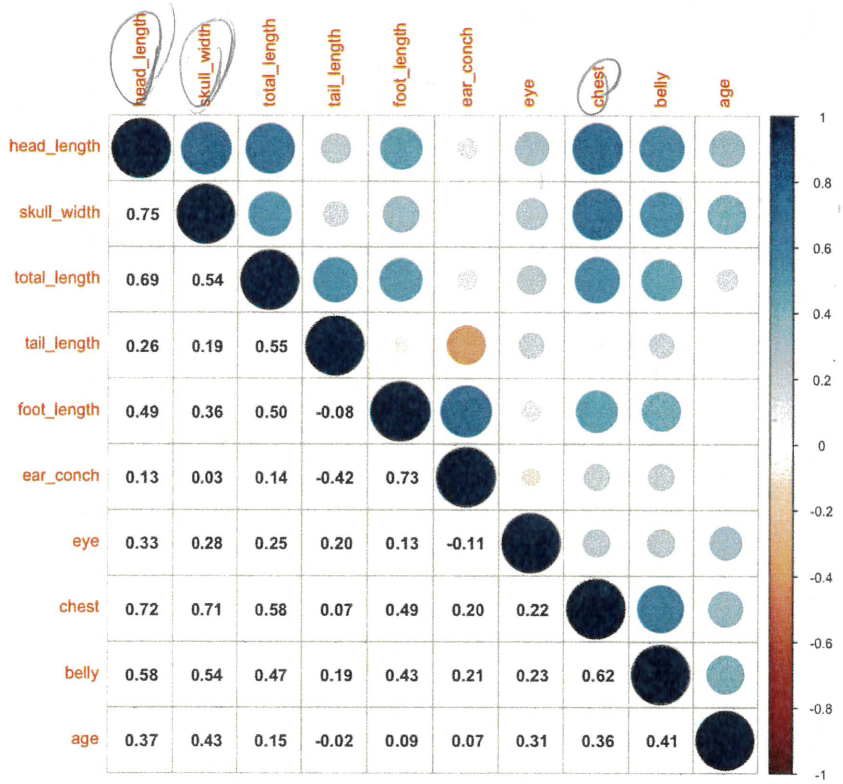
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

| Variable | Description |
|---|---|
| head_length | in mm |
| skull_width | in mm |
| total_length | in cm |
| tail_length | in cm |
| foot_length | in mm |
| ear_conch | ear conch circumference (in mm) |
| eye | width of right eye (in mm) |
| chest | chest girth (in cm) |
| belly | belly girth (in cm) |
| age | in years |

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



**Question 20:**

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.165 \pm (2.01 * .105)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

With repeated sampling, 95% of the time the beta for belly girth would be between -0.023 and 0.443 cm.

**Question 21:** Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The F-test is saying that at least one slope is non-zero, but due to correlated predictors sharing some of the explanatory power for age, not enough unique information is had on those predictors and so their errors are too large and then significance is not obtained.

**Question 22:** Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best subsets goes through every possible model ($2^p$), or in this case, 512 possible models, and keeps the combination of predictors with the lowest AIC value.

**Question 23:** Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-$R^2$ among all possible models.

(A) TRUE

(B) FALSE

It wasn't evaluated on that criterion/metric chosen

**Question 24:** Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM